



GROUP number 03.

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PRIME HORIZON HOTEL

BACKGROUND

Prime Horizon Hotel is a prestigious 4-star luxury hotel located in the heart of Zanzibar, Tanzania. Established in 2015.

The hotel features include:

- 150 Rooms
- 3 Restaurants
- 2 Conference Halls
- Wellness Center with massage and gym facilities
- A Swimming Pool
- 24/7 Room Service

The hotel employs approximately 100 staff members across various departments

VISION AND MISSION

► Vision

Being the premier 5-star hotel in Zanzibar, Tanzania, recognised for excellence in hospitality, innovation, and guest satisfaction.

► Mission

To consistently deliver exceptional guest experiences by providing high-quality accommodation, outstanding customer service, and modern facilities, while creating value for our employees, partners, and the community.

BUSINESS PROCESSES

- Reservation and Booking Process:
- Guest Check-in Process
- Housekeeping Process
- Food and Beverage Service
- Billing and Payment Process
- Reporting and Management Process

EXISTING PROBLEMS

- Double Booking Issues
- Data Inconsistency and Errors
- Inefficient Room Status Management
- Delayed and Inaccurate Reporting
- Revenue Leakage
- Lack of Security and Data Backup
- Poor Decision-Making Support

WHY WE NEED A DATABASE

- Automation of Manual Processes
- Centralised Data Storage
- Real-Time Room Availability
- Improved Revenue Management
- Efficient Reporting and Decision-Making
- Data Security and Backup
- Competitive Advantage

SYSTEM USERS

- System Administrator
- General Manager
- Front Desk Receptionists
- Housekeeping Staff
- Restaurant and Food & Beverage Staff
- Finance and Accounting Staff
- Guests

FUNCTIONAL REQUIREMENTS

- Guest Management
- Room Management
- Reservation Management
- Check-in and Check-out
- Payment and Billing
- Service Management
- Reporting and Analytics
- User Management and Security

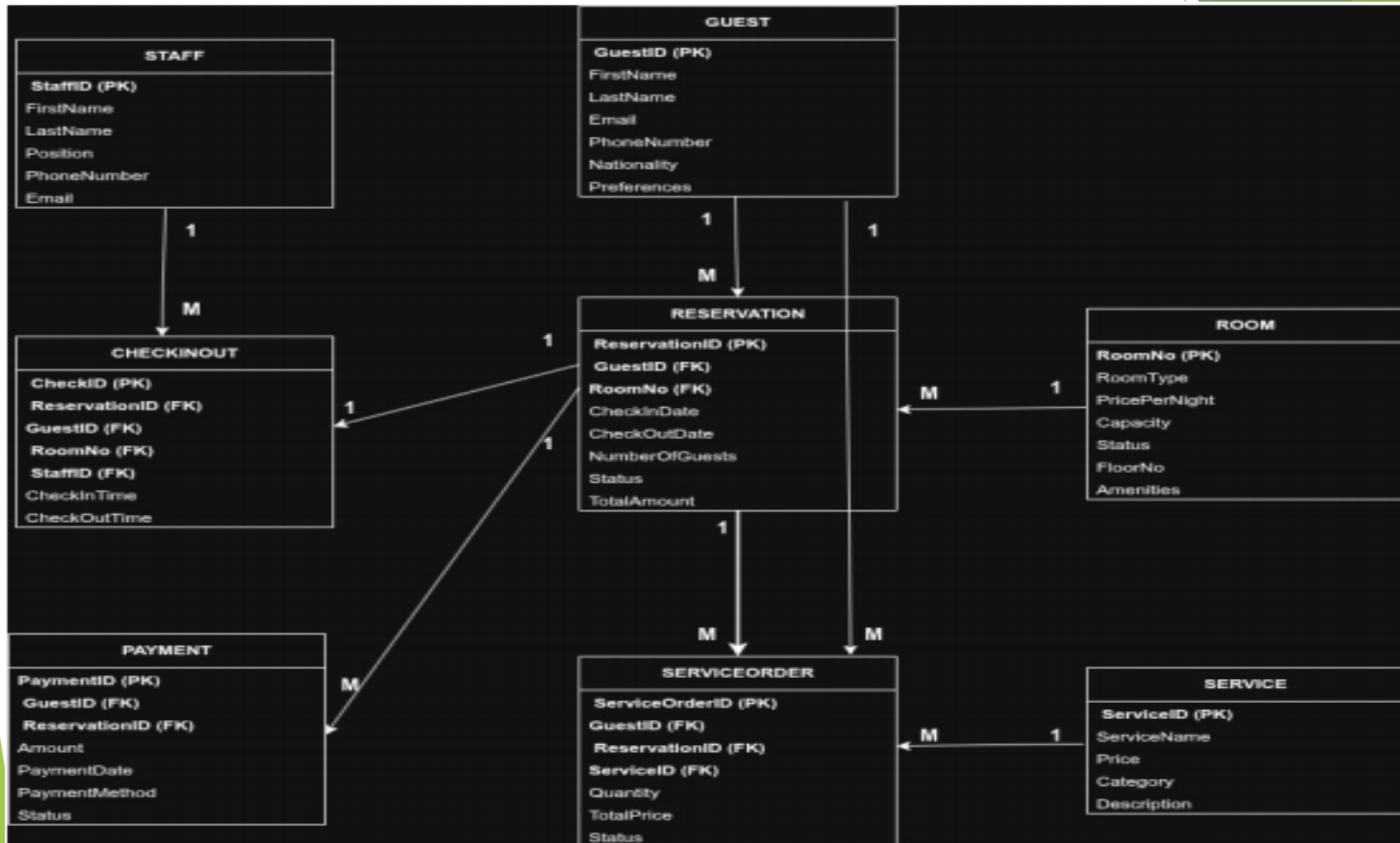
NON-FUNCTIONAL REQUIREMENTS

- Performance Requirements
- Security Requirements
- Reliability and Availability Requirements
- Usability Requirements
- Scalability Requirements
- Maintenance and Support Requirements
- Backup and Recovery Requirements

Database Design: Entities

Entity	Description
Guest	Stores guest personal information
Room	Stores room details and status
Reservation	Stores booking information
CheckInOut	Stores check-in and check-out details
Payment	Stores payment transactions
Service	Stores additional services offered
Service Order	Stores services requested by guests
Staff	Stores hotel employee information

ERD



NORMALIZATION

This section demonstrates the normalization of the Prime Horizon Hotel database from unnormalized Form (UNF) through to Third Normal Form (3NF).

The goal is to ensure that the database design is efficient, consistent, and free from anomalies. Below is example for unnormalized form:

Table: Hotel_Record (UNF)

ResID	GuestName	GuestPhone	GuestEmail	Nationality	RoomNo	RoomType	Price	Floor	CheckIn	CheckOut	Services	SvcPrices
1	John Smith	0712-345-678	john@email.com	USA	101	Standard	\$120	1	10/07/26	15/07/26	Room Service, Laundry	\$15, \$20
1	John Smith	0712-345-678	john@email.com	USA	101	Standard	\$120	1	10/07/26	15/07/26	Room Service, Laundry	\$15, \$20
2	Maria Garcia	0712-345-679	maria@email.com	Spain	202	Deluxe	\$180	2	12/07/26	18/07/26	Spa, Room Service	\$60, \$15
3	Ahmed Hassan	0712-345-680	ahmed@email.com	Tanzania	301	Suite	\$280	3	20/07/26	25/07/26	Airport Transfer	\$40

SQL Queries

Example outputs for Select query:

```
297      -- View all staff members
298 •    SELECT * FROM Staff;
299
300      -- View all services
301 •    SELECT * FROM Service;
302
303      -- View all payments
304 •    SELECT p.PaymentID,
305           CONCAT(g.FirstName, ' ', g.LastName) AS GuestName,
306           p.Amount,
307           p.PaymentMethod,
308           p.PaymentDate,
309           p.Status
310      FROM Payment p
311     JOIN Guest g ON p.GuestID = g.GuestID;
```

REPORTS AND STRATEGIC ANALYSIS

2. Management Reports

Report 1: Monthly Revenue Report

Purpose: Shows hotel revenue performance by month to track financial health.

```
SELECT
    DATE_FORMAT(p.PaymentDate, '%M %Y') AS Month,
    COUNT(p.PaymentID) AS TotalTransactions,
    SUM(p.Amount) AS TotalRevenue,
    AVG(p.Amount) AS AverageTransaction,
    MIN(p.Amount) AS MinPayment,
    MAX(p.Amount) AS MaxPayment,
    COUNT(DISTINCT p.GuestID) AS UniqueGuests
FROM Payment p
WHERE p.Status = 'Success'
GROUP BY DATE_FORMAT(p.PaymentDate, '%Y-%m'),
DATE_FORMAT(p.PaymentDate, '%M %Y')
ORDER BY p.PaymentDate DESC;
```

Sample Output:

Month	TotalTransactions	TotalRevenue	AverageTransaction	UniqueGuests
July 2026	8	\$4,890.00	\$611.25	12
August 2026	2	\$1,660.00	\$830.00	4

1) How these reports help management.

- ***Financial Management***
- ***Room Management***
- ***Guest Management***
- ***Service Management***
- ***Staff Management***

2) How the database supports organizational strategy.

Strategic Area	How the Database Supports It
Revenue Optimization	Tracks all income sources and identifies the highest revenue generators.
Cost Reduction	Identifies inefficiencies, such as empty rooms and low-demand services.
Guest Satisfaction	Stores guest preferences, enabling personalized service.
Competitive Advantage	Data-driven decisions help the hotel outperform competitors.
Marketing Effectiveness	Guest demographics guide targeted campaigns.
Operational Efficiency	Real-time data speeds up processes such as check-in and reservations.
Scalability	The database grows with the hotel and supports future expansion.

3) How managers can use the information for decision-making

1.Strategic Planning Decisions

Data Source	Decision
Room occupancy	Adjust daily room prices (dynamic pricing).
Staff performance	Schedule staff based on occupancy projections.
Service revenue	Run promotions on underperforming services.

2. Tactical Decisions

Data Source	Decision
Revenue reports	Set annual revenue targets and budgets.
Occupancy trends	Plan expansions or renovations during low seasons.
Guest demographics	Enter new markets or launch targeted campaigns.

3. Operational Decisions

Data Source	Decision
Daily operations	Allocate housekeeping based on check-in/out volume.
Payment methods	Offer discounts for preferred payment methods.
Service orders	Ensure popular services are always available.

THANK YOU